Ningkang ZHAO

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Education

Master of Engineering in Energy and Power Engineering
Shanghai Jiao Tong University 2022.09 - 2025.03

Bachelor of Engineering in Electrical Engineering and Automation
Changsha University of Science & Technology 2017.09 - 2021.06

Publications

[1] W. Xu, **N. Zhao**, M. Song, and X. Liu, Surface Modification of Zirconium Alloy with Nanocrystalline Diamond: Enhancing Quenching Performance for Accident Tolerant Fuel. Small 21, no. 40 (2025): e03963. DOI:10.1002/smll.202503963

[2] **N. Zhao**, M. Song, X. Zhang, W. Xu and X. Liu, Nanodiamond Coating in Energy and Engineering Fields: Synthesis Methods, Characteristics, and Applications. Small 2024, 20, 2401292. DOI:10.1002/smll.202401292

[3] W. Xu, L. Tang, **N. Zhao**, K. Ouyang, H. He and X. Liu, Corrosive Effect on Saturated Pool Boiling Heat Transfer Characteristics of Metallic Surfaces with Hierarchical Micro/Nano Structures. Heliyon 2024, 10 (8). DOI:10.1016/j.heliyon.2024.e29750

Patents & Software

[1] High-temperature quenching and high-temperature steam oxidation experimental device. Patent Application No. CN202211454786.5.

[2] Modification method for nuclear fuel cladding based on nanodiamond coatings. Patent Application No. CN202311253006.5.

[3] Fuel Cladding Film Boiling Flow Pattern Recognition and Key Parameter Calculation [MAST v1.0]. Software Copyright.

Research Experience

Research Assistant Shanghai Jiao Tong University 2025.09 - 2026.06

- Performed first-principles calculations using **Quantum ESPRESSO**, including bulk and surface model construction, k-point and cutoff-energy convergence testing, structural relaxation, and surface-related calculations.
- Assisted with molecular-dynamics studies of zirconium-alloy oxidation, including model design, parameter assessment, convergence evaluation, and interpretation of atomistic transport behavior.
- Proposed an **oxygen-impedance model** to describe coupled oxygen transport and reaction across multilayer coating/cladding systems and to bridge atomistic transport parameters with macroscopic oxidation kinetics.

Master's thesis

Accident Tolerant Mechanism of Zirconium Alloy Fuel Cladding Based On Nanodiamond Coating

- Investigated the high-temperature **quenching and oxidation behavior** of Zr alloy cladding modified with nanocrystalline diamond coatings for accident-tolerant fuel applications.
- Designed and constructed experimental systems for **high-temperature quenching and steam oxidation**, integrating temperature acquisition and high-speed visualization.
- Characterized coating morphology and structural evolution using **SEM/EDS, AFM, XPS, and Raman spectroscopy**, combined with MATLAB-based image and transient boiling analysis.
- Demonstrated a **21.3% shorter film-boiling stage**, a **15.06% higher boiling-transition temperature**, and a **17.7% increase in peak cooling rate** for NCD-coated Zr.

Selected Projects

2024 Joint Summer School on Nuclear Science and Technology Kyushu University, Japan

- Participated in the conceptual design and thermal-hydraulic evaluation of a lead-bismuth-cooled natural circulation reactor.

Artificial Intelligence Competition

Intelligent Identification and Conversion of Nuclear Power Plant System Flow Charts

- Developed and optimized a **YOLOv10**-based object-detection model for automated recognition of components in nuclear power plant system diagrams.

Mathematical Modeling Competition

Intelligent Clinical Diagnosis and Treatment Modeling of Hemorrhagic Stroke

- Developed **BP neural-network and random-forest models** for multi-parameter clinical-data analysis and outcome prediction.

Awards & Honors

- **Second-class Corporate Scholarship**, Shanghai Jiao Tong University, China.
- **Silver Medal** in the 2024 Joint Summer School, Kyushu University, Japan.
- **Second Prize** in "Huawei Cup" China Graduate Mathematical Modeling Competition, Shanghai Jiao Tong University, China.

Skills

- **Computational:** Quantum ESPRESSO, Python, MATLAB, C; ASE and VESTA (basic).
- **Experimental:** High-temperature quenching/steam oxidation; SEM/EDS, AFM, XPS, Raman; high-speed imaging.
- **Languages:** Chinese (native), English.